

Practice Questions from Week 5

Question 5 : 640653770609

Total Mark : 0.00 | Type : COMPREHENSION

Consider a regression problem where you are tasked with predicting the sale prices of houses based on their square footage. You decide to experiment with two different models: The training dataset consists of information on 200 houses, and you use the models to make predictions on a test dataset of 50 houses. The Mean Squared Error (MSE) is chosen as the evaluation metric. Based on the above data, answer the given subquestions.

Model P : $\hat{y}_i = w_0 + w_1x$
Model Q : $\hat{y}_i = w_0 + w_1x + w_2x^2$



Discussions (0)



Question 6 : 640653770610

View Parent QN

View Solutions (1)

Total Mark : 3.00 | Type : MCQ

Considering the specific context of predicting house prices based on square footage, which model is more likely to provide accurate predictions on the training dataset?

OPTIONS :

- ☐ Model P
- ☐ Model Q
- ☐ Both models are equally likely to provide accurate predictions
- ☐ It depends on the distribution of house prices in the dataset

Solution

Model P is a simpler linear model with only two parameters (w_0 , w_1) and Model Q is more complex, with three parameters (w_0 , w_1 , w_2). A complex model (Model Q) generally has greater flexibility and can fit the training data more closely compared to a simpler model (Model P). This is because Model Q can capture both linear and quadratic patterns in the data, whereas Model P can only capture linear trends. Therefore, Model Q is more likely to provide accurate predictions on the training dataset because it has greater flexibility to fit the data.

Question 7 : 640653770611

 View Parent QN

 View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Identify the factors that could influence the model's performance on the training dataset in this housing price prediction scenario. Select all correct statements:

OPTIONS :

- ☐ Model P may struggle to capture non-linear relationships present in houseprice data.
- ☐ Model Q might be sensitive to outliers in the square footage variable.
- ☐ The choice between Model P and Model Q depends on the budget constraints of potential homebuyers.
- ☐ Model Q will always perform well on the test dataset.

Solution

a) Model P is a linear regression model and cannot account for non-linear patterns in the data. If house prices have a non-linear relationship with square footage, Model P will struggle to fit the training data accurately.

b) Model Q includes a quadratic term (x^2), which can amplify the effect of outliers in the square footage variable. This makes it more sensitive to outliers compared to Model P.

c) The choice of model depends on the ability to capture relationships in the data, not on external factors like budget constraints of homebuyers. This statement is irrelevant to model performance.

d) While Model Q may fit the training data better due to its flexibility, overfitting could cause poor generalization to unseen test data. It is not guaranteed that Model Q will always perform well on the test dataset.

Question 14 : 640653770613[View Solutions \(0\)](#) Total Mark : 3.00 | Type : SA

Kernel regression with a polynomial kernel of degree three is applied on a data set $\{X, y\}$.
Let the weight vector be given by

$$w = \phi(X)[2.3, -1.0, 0.4, -0.7]^T$$

Here $\phi(X)$ is the transformed data matrix whose i^{th} column is $\phi(x_i)$. What will be the prediction for the data point $[0, 0, 0, 0]^T$?

Answer (Numeric):

Answer

Solution

The prediction of test data point in kernel regression is

$$\sum_{i=1}^n \alpha^* k(x_i, x_{test})$$

The given kernel function is

$$k(x_i, x_{test_i}) = (x^T x_{test_i} + 1)^3$$

$$\phi(x) = [1 \ 1 \ 1 \ 1]$$

$$\sum_{i=1}^n \alpha^* k(x_i, x_{test}) = [1 \ 1 \ 1 \ 1] \begin{bmatrix} 2.3 \\ -1.0 \\ 0.4 \\ -0.7 \end{bmatrix} = 2.3 - 1 + 0.4 - 0.7 = 1$$

Question 16 : 640653770612

[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : MCQ

Let X be the data matrix of shape (d, n) and y be the corresponding label vector. A linear regression model of the form $\hat{y}_i = w^T x_i$ is fit using the squared error on the same dataset. If the solution w^* to the optimization problem is orthogonal to the subspace spanned of the data point (columns of matrix X), what will be the squared error?

OPTIONS :

- ☐ 0
- ☐ 1
- ☐ $\|y\|^2$
- ☐ Insufficient information to answer

Solution

We are given a data matrix X of shape (d, n) and a corresponding label vector y . A linear regression model of the form

$$\hat{y} = w^T x$$

is fit using the squared error on the same dataset. The solution w^* to the optimization problem is orthogonal to the subspace spanned by the data points (columns of matrix X). We need to determine the squared error. In linear regression, the optimal weight vector w^* is typically found by minimizing the squared error:

$$\min_w \|y - X^T w\|^2$$

here, $X^T w$ represents the projection of w onto the column space of X . This implies that the projection of y onto the column space of X is zero

$$\|y - 0\|^2$$

$$\|y\|^2$$

Question 3 : 640653852824[View Solutions \(0\)](#) Total Mark : 3.00 | Type : SA

Consider the following training dataset for a linear regression problem with one feature:

x_i	y_i
1	1.2
-1	0.1
2	2.1
-2	-1

Find the value of w^* , the optimal weight. _____

Answer (Numeric):

Answer

Solution

$$w^* = (XX^T)^\dagger (Xy)$$

$$X = \begin{bmatrix} 1 & -1 & 2 & -2 \end{bmatrix} \quad y = \begin{bmatrix} 1.2 \\ 0.1 \\ 2.1 \\ -1 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 \\ -1 \\ 2 \\ -2 \end{bmatrix}$$

$$XX^T = 10$$

$$(XX^T)^\dagger = \frac{1}{10}$$

$$Xy = 7.3$$

$$w^* = (XX^T)^\dagger (Xy) = \frac{1}{10} \times 7.3 = 0.73$$

Question 4 : 640653852825

[View Solutions \(0\)](#) Total Mark : 3.00 | Type : SA

Consider the following dataset for a kernel regression problem with a polynomial kernel of degree two along with the coefficient vector α :

$$X = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}, y = \begin{bmatrix} 1 \\ -1 \\ 2 \\ -2 \end{bmatrix}, \alpha = \begin{bmatrix} \alpha_1 \\ -0.625 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

If the prediction for the data-point $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is 0, find the value of α_1 . _____

Answer (Numeric):

Answer

Solution

The prediction of test data point in kernel regression is

$$\sum_{i=1}^n \alpha^* k(x_i, x_{test})$$

The given kernel function is

$$\begin{aligned} k(x_i, x_{test_i}) &= (x^T x_{test_i} + 1)^2 \\ k\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) &= \left(\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1\right)^2 = 4 \\ k\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) &= \left(\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1\right)^2 = 4 \\ k\left(\begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) &= \left(\begin{bmatrix} -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1\right)^2 = 0 \\ k\left(\begin{bmatrix} 0 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) &= \left(\begin{bmatrix} 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1\right)^2 = 0 \end{aligned}$$

$$\sum_{i=1}^n \alpha^* k(x_i, x_{test}) = 0$$

$$\Rightarrow \alpha_1(4) + \alpha_2(4) + \alpha_3(0) + \alpha_4(0) = 0$$

$$\Rightarrow \alpha_1(4) - 0.625 \times 4 = 0$$

$$\Rightarrow \alpha_1(4) = 0.625 \times 4$$

$$\Rightarrow \alpha_1 = \frac{0.625 \times 4}{4}$$

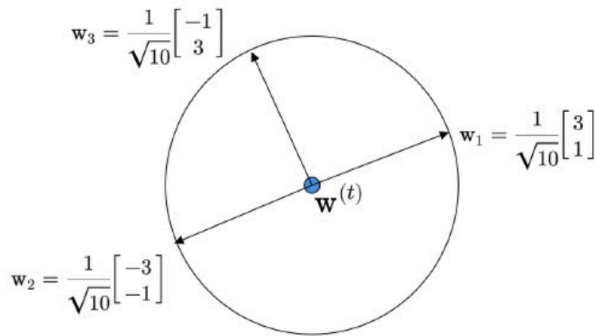
$$\Rightarrow \alpha_1 = 0.625$$

Question 6 : 640653852827

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider a linear regression problem that has two features with a training dataset of 10,000 points. The blue point in the image below is the weight vector, $\mathbf{w}^{(t)}$, at some iteration of gradient descent. $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ are unit vectors. Specifically, $\mathbf{w}_1$ is a unit vector in the direction of the gradient at this point.



Question 7 : 640653852828

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If the next iteration of gradient descent is expressed as $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} + \eta \mathbf{w}_k$ with $\eta > 0$, what would be the value of k ? _____

Answer (Numeric):

Answer

[Check Answer](#)

Solution

2 (Visual Inspection)

Question 8 : 640653852829

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If $\mathbf{w}^{(t)} = \begin{bmatrix} 0.4 \\ 0.1 \end{bmatrix}$ and $\eta = \frac{1}{\sqrt{10}}$, find $\|\mathbf{w}^{(t+1)}\|$,



where we use the usual L_2 norm. _____

Answer (Numeric):

Answer

Solution

$$\begin{aligned} w^{(t+1)} &= w^t + \eta w_k^t \\ &= \begin{bmatrix} 0.4 \\ 0.1 \end{bmatrix} + \frac{1}{\sqrt{10}} \times \left(\frac{1}{\sqrt{10}} \right) \begin{bmatrix} -3 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 0.4 \\ 0.1 \end{bmatrix} + \begin{bmatrix} -0.3 \\ -0.1 \end{bmatrix} \\ &= \begin{bmatrix} 0.1 \\ 0 \end{bmatrix} \end{aligned}$$

$$\therefore \left\| \begin{bmatrix} 0.1 \\ 0 \end{bmatrix} \right\|_2 = \sqrt{(0.1)^2 + (0)^2} = 0.1$$

Question 2 : 6406531021341

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Kernel regression with a polynomial kernel is applied on the following dataset with two features:

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad y = [2, 1, 2]^T$$

Weight vector can be written as $w = \phi(X)\alpha$, where ϕ is the transformation mapping corresponding to the kernel

$k(x_i, x_j) = (1 + x_i^T x_j)^2$. The vector α is given by $(K)^{-1}y$,

where K is the kernel matrix.

Question 3 : 6406531021342

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : MCQ

Compute K .

OPTIONS :

☐ $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 4 & 1 \\ 4 & 1 & 1 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 4 & 1 \\ 1 & 1 & 1 \\ 4 & 1 & 1 \end{pmatrix}$

☐ $\begin{pmatrix} 4 & 4 & 1 \\ 1 & 1 & 1 \\ 4 & 1 & 1 \end{pmatrix}$

☐ $\begin{pmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

Solution

Given $k(x_i, x_j) = (1 + x_i^T x_j)^2$

$$\therefore k\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \left(1 + \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)^2 = (1 + 1)^2 = 2^2 = 4$$

$$k\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \left(1 + \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)^2 = 1$$

$$k\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = \left(1 + \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right)^2 = 1$$

$$\therefore k = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Question 4 : 6406531021343

[View Parent QN](#)

[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : SA

What will be the prediction for the data point $[1, -1]^T$? Enter the answer correct to two decimal places.

Answer (Numeric):

Answer

Solution

We Know, $\alpha^* = K^{-1}y$

$$K = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad M = \begin{bmatrix} 3 & 0 & -3 \\ 0 & 3 & 3 \\ -3 & 3 & 15 \end{bmatrix} \quad C = \begin{bmatrix} 3 & 0 & -3 \\ 0 & 3 & -3 \\ -3 & -3 & 15 \end{bmatrix} \quad C^T = Adj = \begin{bmatrix} 3 & 0 & -3 \\ 0 & 3 & -3 \\ -3 & -3 & 15 \end{bmatrix}$$

$$K^{-1} = \frac{Adj}{\det(K)} = \frac{1}{9} \begin{bmatrix} 3 & 0 & -3 \\ 0 & 3 & -3 \\ -3 & -3 & 15 \end{bmatrix} \quad K^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{5}{3} \end{bmatrix}$$

$$K^{-1}y = \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{5}{3} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{3} \\ \frac{7}{3} \end{bmatrix}$$

$$\text{Given } k(x_i, x_j) = (1 + x_i^T x_j)^2$$

$$\therefore k\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \left(1 + \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)^2 = (1 + 1)^2 = 2^2 = 4$$

$$k\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \left(1 + \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)^2 = (1 - 1)^2 = 0$$

$$k\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \left(1 + \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)^2 = 1^2 = 1$$

$$K = \begin{bmatrix} 4 & 0 & 1 \end{bmatrix}$$

$$\text{We Know, } \sum_{i=1}^n \alpha^* k(x_i, x_{test})$$

$$= \sum_{i=1}^n \begin{bmatrix} 0 \\ -1 \\ 3 \\ 7 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 0 & 1 \end{bmatrix} = \frac{7}{3} \approx 2.33$$

Question 21 : 6406531021344

[View Solutions \(0\)](#) Total Mark : 3.00 | Type : MSQ

Consider a linear regression model with loss $L(w) = \sum_{i=1}^n (w^T x_i - y_i)^2$, where x_i 's are the d -dimensional training data points and y_i 's are their corresponding labels. For the optimal weight vector w^* , which among the following is correct?

OPTIONS :

- ☐ If we double all the values of y_i , w^* will also get doubled. 
- ☐ If we double all the values of y_i , w^* will get halved. 
- ☐ If we double all the values of x_i , w^* will get halved. 
- ☐ If we double all the values of x_i , w^* will also get doubled. 

Solution

(a) , (c)

$$\text{We Know, } L(w) = \sum_{i=1}^n (w^T x_i - y_i)^2$$

$$\therefore w^* = (XX^T)^+ Xy$$

If we double all the values of y_i

$$w' = (XX^T)^+ X(2y)$$

$$w' = 2((XX^T)^+ Xy)$$

$$w' = 2w^*$$

if we double all the value of x_i

$$w' = ((2X)(2X^T))^+ (2X)y$$

$$w' = (4XX^T)^+ (2X)y$$

$$w' = \frac{1}{4} (XX^T)^+ (2X)y$$

$$w' = \frac{1}{4} \times 2((XX^T)^+ Xy)$$

$$w' = \frac{1}{2} ((XX^T)^+ Xy)$$

$$w' = \frac{1}{2} w^*$$